Do Agents Look for Truth or Comfort? Measuring the Truth Discount in LLM Agent Societies

Anonymous submission

Abstract

1 Introduction
2 Related Work
3 Methodology
4 Experiments
5 Results
6 Discussion

Ethical Statement

This work investigates information diffusion, epistemic verification, and the tension between empirical truth and social deference in LLM agent societies. All simulations were conducted entirely *in silico* within synthetic multi-agent environments; no human participants were involved, deceived, or subjected to misinformation. To proactively prevent dual-use risks and avoid producing actionable disinformation strategies, our experimental design purposefully evaluates benign, low-stakes cultural scenarios (fictitious community concert announcements featuring invented artists) rather than sensitive electoral, biomedical, or financial domains. As autonomous LLM agents are increasingly deployed in decentralized communication, digital governance, and collective consensus platforms, diagnosing vulnerabilities where agents sacrifice factual accuracy for interpersonal comfort or institutional deference is essential for engineering principled auditing tools, epistemic guardrails, and automated verification mechanisms.

References

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A Experimental Protocol and Simulation Details
B Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this manuscript, the authors utilized generative AI assistants (including Google Gemini and Anthropic Claude) to assist with literature search formulation, LaTeX typesetting and structural formatting, editorial refinement of prose, and boilerplate scaffolding for simulation execution and analysis scripts. All research hypotheses,

counterbalanced twin-world experimental designs, formal metric formulations (including the Truth Discount estimators), empirical simulations, and analytical conclusions were conceived, conducted, and rigorously verified by the authors. The authors retain full and sole scientific accountability for the integrity, validity, and original contributions of this research.

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